OPERATING SYSTEM ASSIGNMENT 2

```
rvu@rvu-OptiPlex-SFF-7020: ~/wruthambara
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ nano process_control.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ bash process_control.sh
Parent Control Demonstraction
Parent shell PID:8232
starting a long-running process(sleep 100)...
Process started with PID:
Current Jobs:
[1]+  Running                 sleep 100 &
Now you can:
1. User 'fg%1' to bring it to foreground
2. Press ctrl+z to suspend
3. Use 'bg%1' to resume in background
4. Use 'kill' to terminate
^Z
[2]+  Stopped                 bash process_control.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ ps -f --forest
UID          PID    PPID  C STIME TTY          TIME CMD
rvu         4975    4967  0 20:32 pts/0    00:00:00 bash
rvu         8170    4975 69 21:28 pts/0    00:01:24  \_ bash backgound_task.sh
rvu         8232    4975  0 21:30 pts/0    00:00:00  \_ bash process_control.sh
rvu         8233    8232  0 21:30 pts/0    00:00:00  |   \_ sleep 100
rvu         8251    4975  0 21:30 pts/0    00:00:00  \_ ps -f --forest
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ kill 4975
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ nano backgound_task.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ bash backgound_task.sh
Background Task  Started
Process ID:8509
Running..Count=1
^Z
[3]+  Stopped                 bash backgound_task.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ jobs
[1]   Stopped                 bash backgound_task.sh
[2]-  Stopped                 bash process_control.sh
[3]+  Stopped                 bash backgound_task.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ ps -ef --forest
```

```
GNU nano 7.2                                   process_control.sh *
#!/bin/bash
echo "Parent Control Demonstraction"
echo "Parent shell PID:$$"
echo "starting a long-running process(sleep 100)..."
sleep 100 &
PID1=$!
echo "Process started with PID:$PID1"
echo "Current Jobs:"
jobs
echo "Now you can:"
echo "1. User 'fg%1' to bring it to foreground"
echo "2. Press ctrl+z to suspend"
echo "3. Use 'bg%1' to resume in background"
echo "4. Use 'kill$PID1' to terminate"
wait

^G Help      ^O Write Out   ^W Where Is   ^K Cut       ^T Execute   ^C Location   M-U Undo      M-A Set Mark   M-] To Bracket
^X Exit      ^R Read File   ^\ Replace    ^U Paste     ^J Justify   ^/ Go To Line  M-E Redo      M-6 Copy       ^Q Where Was
```

```
rvu@rvu-OptiPlex-SFF-7020: ~/wruthambara
GNU nano 7.2                                   backgound_task.sh *
#!/bin/bash
echo "Background Task  Started"
echo "Process ID:$$"
count=1
while true
do
        echo "Running..Count=$count "
        count=$((count))
        sleep 5
done
```

```
rvu@rvu-OptiPlex-SFF-7020: ~/wruthambara
Process ID:8369
Running..Count=1
^Z
[3]+  Stopped                 bash backgound_task.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ jobs
[1]   Stopped                 bash backgound_task.sh
[2]-  Stopped                 bash process_control.sh
[3]+  Stopped                 bash backgound_task.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ ps -ef --forest
UID          PID    PPID  C STIME TTY          TIME CMD
root           2       0  0 20:30 ?        00:00:00 [kthreadd]
root           3       2  0 20:30 ?        00:00:00  \_ [pool_workqueue_release]
root           4       2  0 20:30 ?        00:00:00  \_ [kworker/R-rcu_gp]
root           5       2  0 20:30 ?        00:00:00  \_ [kworker/R-sync_wq]
root           6       2  0 20:30 ?        00:00:00  \_ [kworker/R-kvfree_rcu_reclaim]
root           7       2  0 20:30 ?        00:00:00  \_ [kworker/R-slub_flushwq]
root           8       2  0 20:30 ?        00:00:00  \_ [kworker/R-netns]
root          10       2  0 20:30 ?        00:00:00  \_ [kworker/0:0H-events_highpri]
root          13       2  0 20:30 ?        00:00:00  \_ [kworker/R-mm_percpu_wq]
root          14       2  0 20:30 ?        00:00:00  \_ [rcu_tasks_kthread]
root          15       2  0 20:30 ?        00:00:00  \_ [rcu_tasks_rude_kthread]
root          16       2  0 20:30 ?        00:00:00  \_ [rcu_tasks_trace_kthread]
root          17       2  0 20:30 ?        00:00:00  \_ [ksoftirqd/0]
root          18       2  0 20:30 ?        00:00:02  \_ [rcu_preempt]
root          19       2  0 20:30 ?        00:00:00  \_ [rcu_exp_par_gp_kthread_worker/1]
root          20       2  0 20:30 ?        00:00:00  \_ [rcu_exp_gp_kthread_worker]
root          21       2  0 20:30 ?        00:00:00  \_ [migration/0]
root          22       2  0 20:30 ?        00:00:00  \_ [idle_inject/0]
root          23       2  0 20:30 ?        00:00:00  \_ [cpuhp/0]
root          24       2  0 20:30 ?        00:00:00  \_ [cpuhp/2]
root          25       2  0 20:30 ?        00:00:00  \_ [idle_inject/2]
root          26       2  0 20:30 ?        00:00:00  \_ [migration/2]
root          27       2  0 20:30 ?        00:00:00  \_ [ksoftirqd/2]
root          28       2  0 20:30 ?        00:00:00  \_ [kworker/2:0-mm_percpu_wq]
root          29       2  0 20:30 ?        00:00:00  \_ [kworker/2:0H-events_highpri]
root          30       2  0 20:30 ?        00:00:00  \_ [cpuhp/4]
root          31       2  0 20:30 ?        00:00:00  \_ [idle_inject/4]
root          32       2  0 20:30 ?        00:00:00  \_ [migration/4]
```

```
rvu         8170    4975 23 21:28 pts/0    00:01:24  |       \_ bash backgound_task.sh
rvu         8232    4975  0 21:30 pts/0    00:00:00  |       \_ bash process_control.sh
rvu         8233    8232  0 21:30 pts/0    00:00:00  |       |   \_ sleep 100
rvu         8509    4975  0 21:33 pts/0    00:00:00  |       \_ bash backgound_task.sh
rvu         8511    8509  0 21:33 pts/0    00:00:00  |       |   \_ sleep 5
rvu         8619    4975  0 21:34 pts/0    00:00:00  |       \_ ps -ef --forest
rvu         6051    2140  0 21:01 ?        00:00:07  \_ /usr/bin/nautilus --gapplication-service
rvu         6109    2140  0 21:01 ?        00:00:02  \_ /usr/bin/eog /home/rvu/Pictures/Screenshots/Screenshot from 2026-02-17 20-57-20.png
root        6024       1  0 21:00 ?        00:00:01 /usr/libexec/fwupd/fwupd
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$ fg % 1
bash backgound_task.sh
Running..Count=1
^Z
[3]+  Stopped                 bash backgound_task.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/wruthambara$
```

nano filename.sh -> Creates Shell Script
Bash -> Parent process
cd -> change directories
ps -> Display process Status & if is a child process
pstree -> Parent child relationship
top -> Shows Table of Process
job -> list of background processes
fg %1 -> Brings job to foreground
bg %1 -> Resumes job in background
Kill -> Terminates process
Sleep 30: Pause processes for 30 Seconds
Sleep 60 &: Run the process in background
ps -f: Full format listing
ps -ef: Shows every process on the System
ps -ef --forest: list of all running process in hierarchical tree
echo $$: Displays process ID (PID)
Sleep 100 &: Sleeps / pause for 100 Seconds
ps -f --forest: Display current Running file